Web Technologies (CSC336) - Lab Mid Term

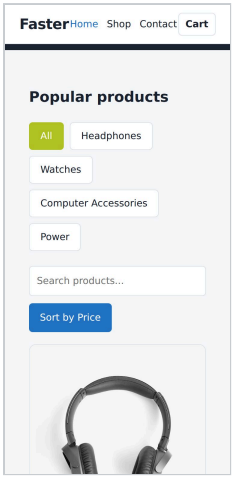
Name: Ahmad **Registration No.:** FA24-BCS-061
COMSATS University Islamabad, Lahore Campus · FA26 BCS A · Instructor: Muhammad Usman Akram

N	BP	H	K	Sticky top	Premium limit
61	661 px	67 (hsl(67, 70%, 45%))	4	(61 mod 40) + 70 = 91 px	61 x 100 = Rs. 6,100

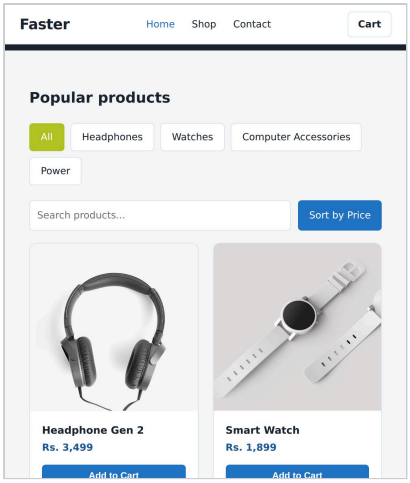
Screenshots were taken in a real browser at 375px, exactly BP (661px) and 1280px. Tall pages are cropped to the top part of the products section; image widths are roughly proportional to the viewport widths.

Screenshots - Task 1: CSS-only category filter

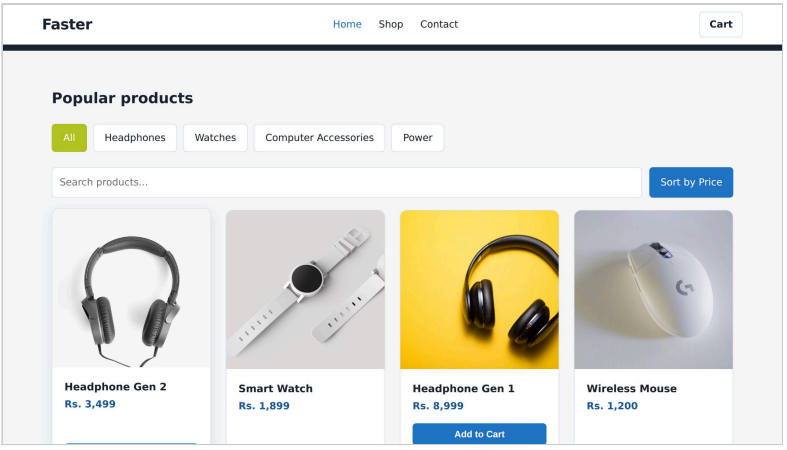
Page load: "All" selected, all 8 cards shown



375px

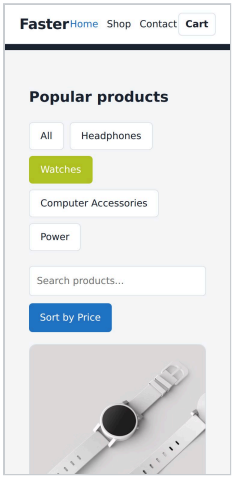


661px (BP)

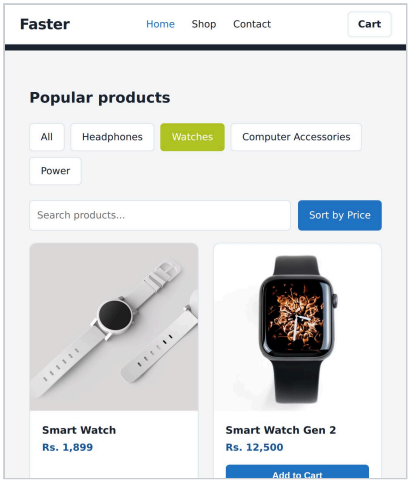


1280px

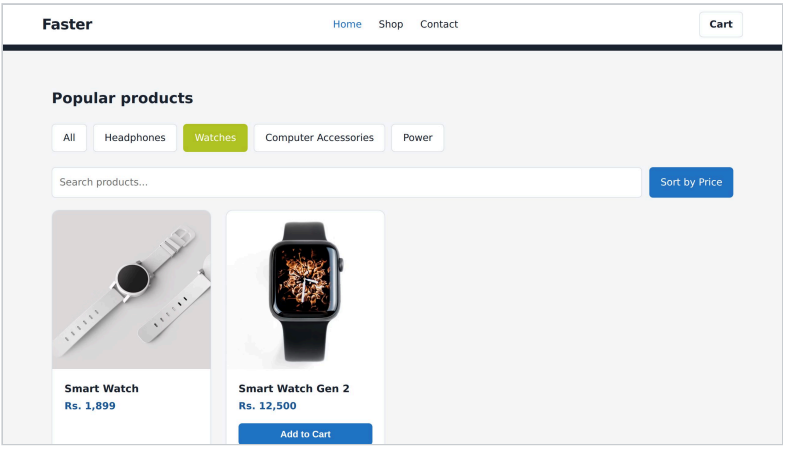
Filter: Watches (2 cards, no gaps)



375px

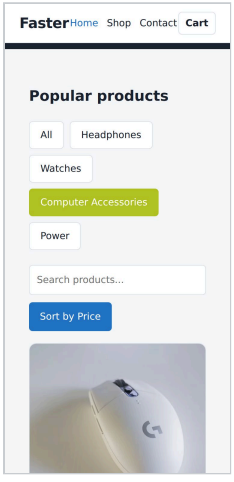


661px (BP)

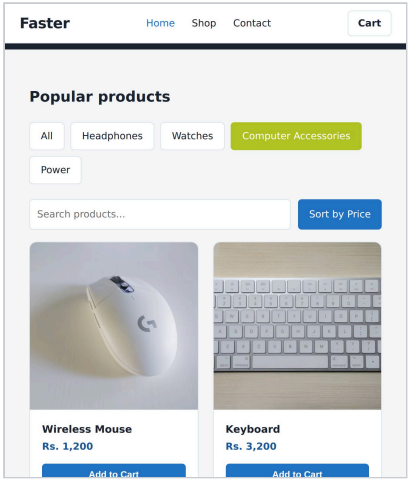


1280px

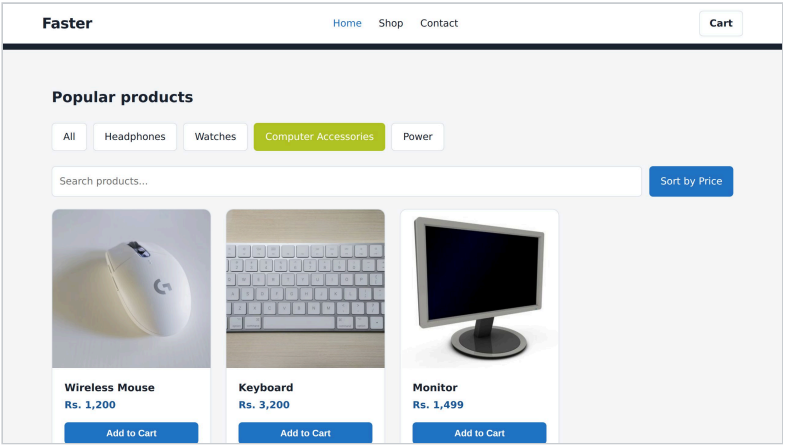
Filter: Computer Accessories (3 cards, no gaps)



375px

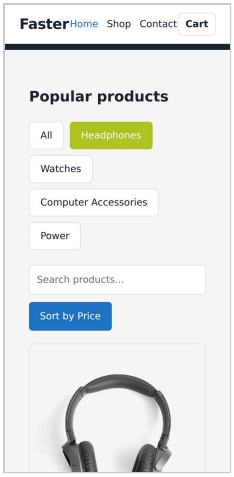


661px (BP)

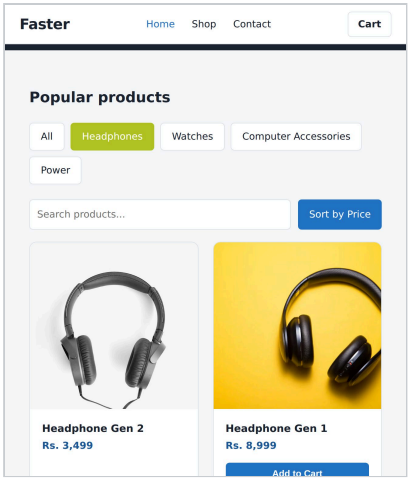


1280px

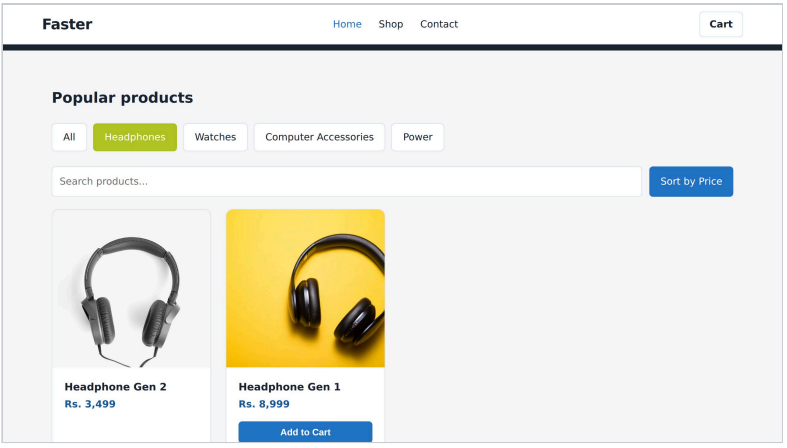
Filter: Headphones (2 cards, no gaps)



375px

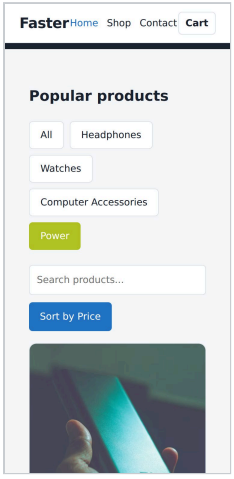


661px (BP)

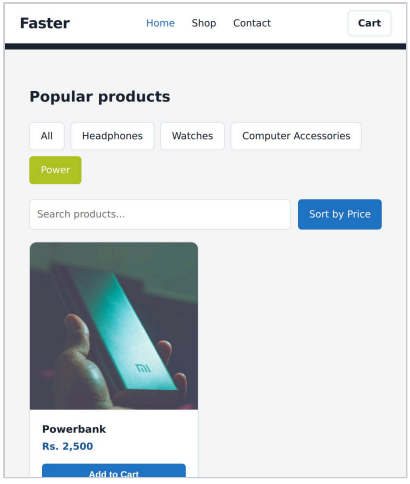


1280px

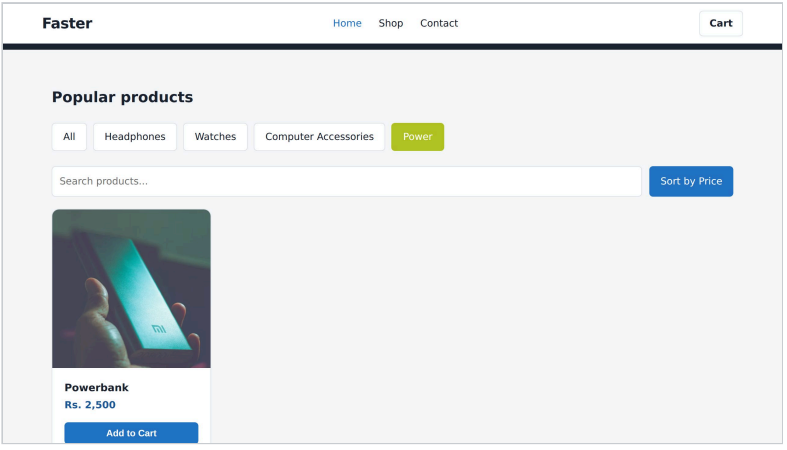
Filter: Power (1 card, no gaps)



375px



661px (BP)



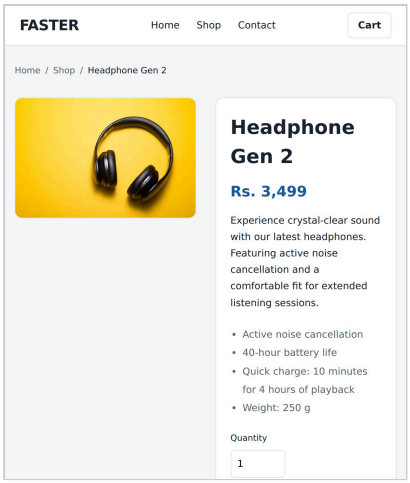
1280px

Screenshots - Task 2: Product page grid + sticky details box

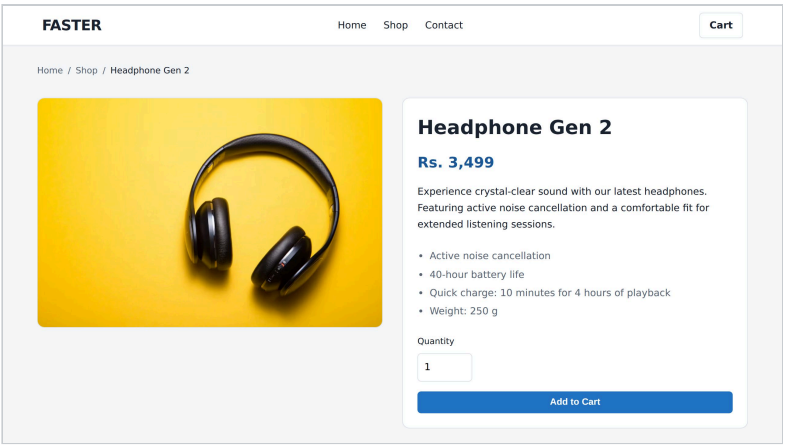
Top of page (below BP: stacked; from BP: side by side, description underneath)



375px

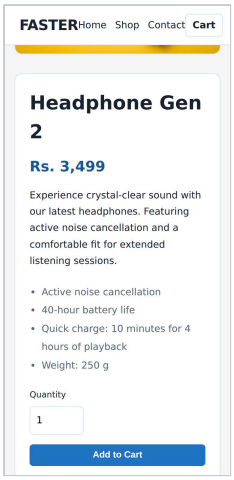


661px (BP)

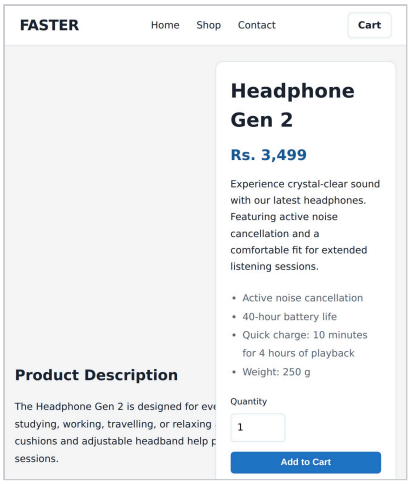


1280px

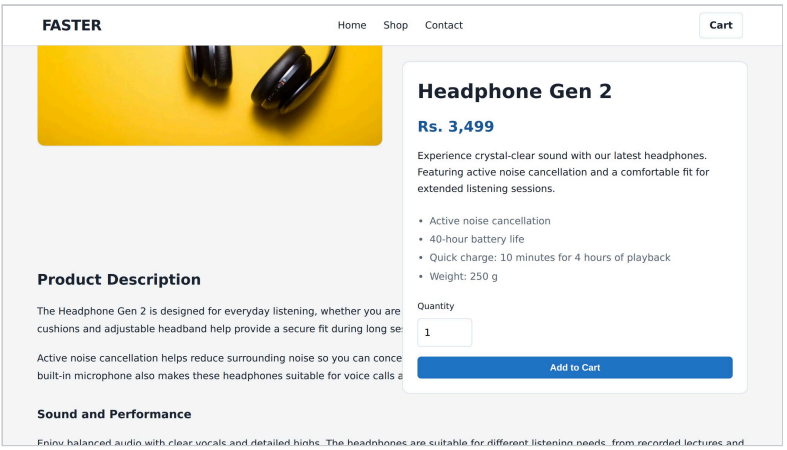
Scrolled 300px (from BP the details box stays at 91px from the top; below BP it scrolls normally)



375px



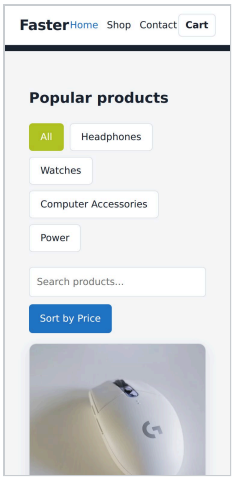
661px (BP)



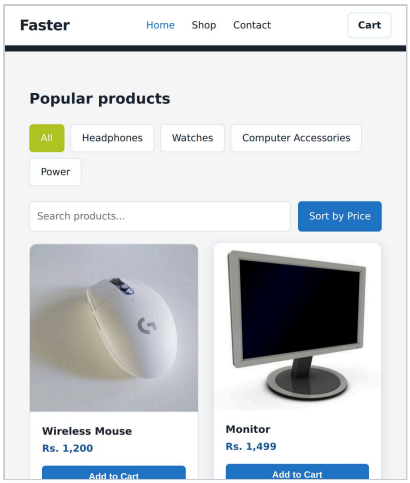
1280px

Screenshots - Task 3: Live search and price sort

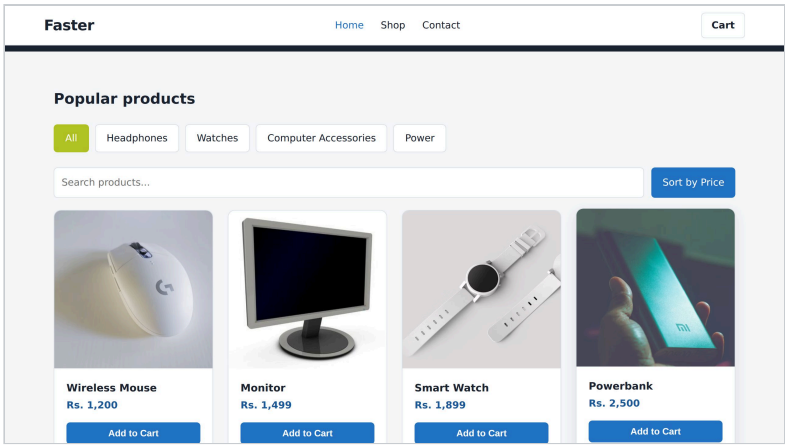
After the 1st click on Sort by price (low to high)



375px

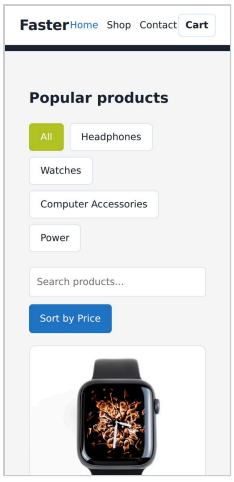


661px (BP)

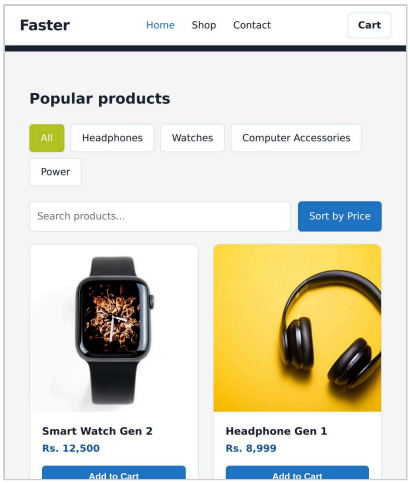


1280px

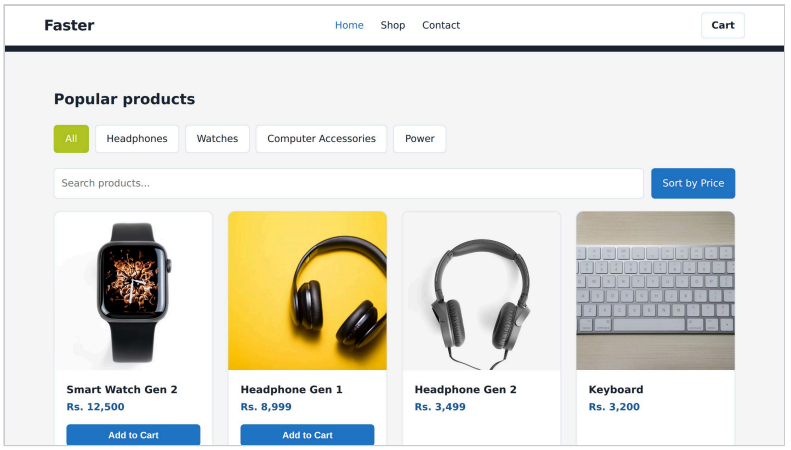
After the 2nd click (high to low)



375px

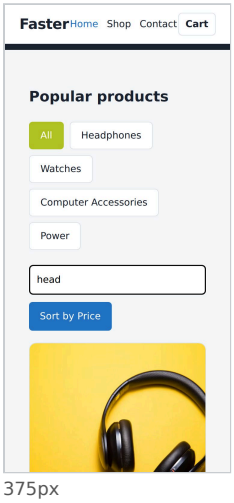


661px (BP)

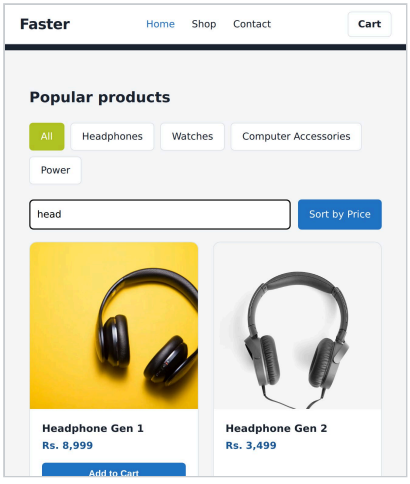


1280px

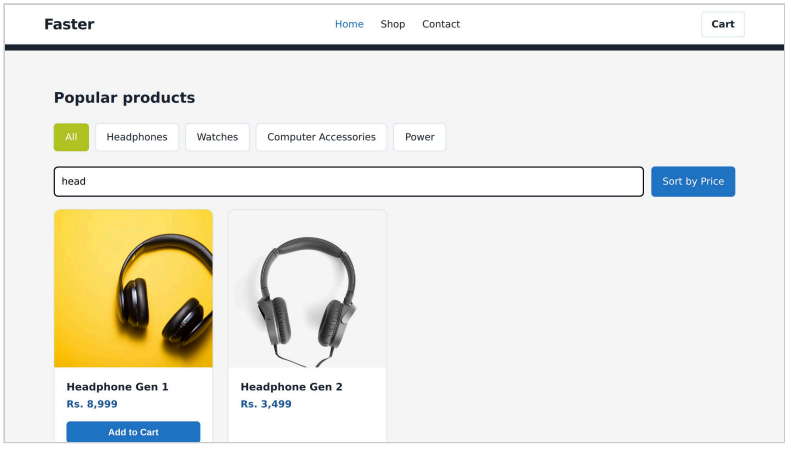
Search "head" (Premium labels visible on Headphone Gen 1 / Smart Watch Gen 2 in the full grid)



375px

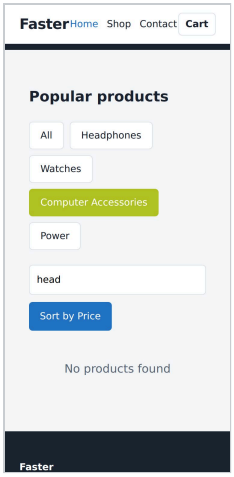


661px (BP)

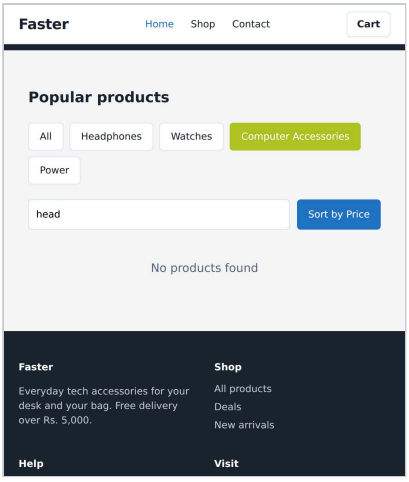


1280px

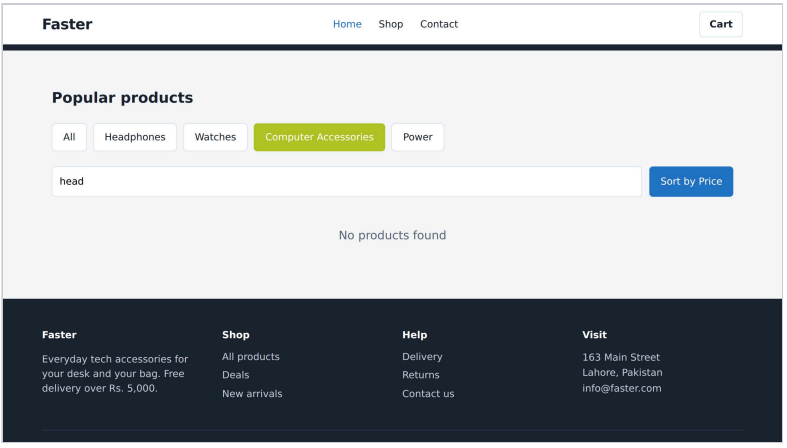
Search "head" + category Computer Accessories = no match: "No products found"



375px



661px (BP)



1280px

Code - Task 1: CSS-only category filter

index.html - category class on each card (example; every card has one: headphones, watches, accessories, power)

```
<article class="product-card headphones">
          <a href="product.html"><img src="./images/1.jpg" alt="Headphone Gen 2" loading="lazy"></a>
          <div class="card-details">
            <h3 class="product-name"><a href="product.html">Headphone Gen 2</a></h3>
            <p class="product-price">Rs. 3,499</p>
            <button type="button" class="btn btn-cart">Add to Cart</button>
          </div>
        </article>
```

index.html - radio buttons and filter labels (added above the grid)

```
<div class="category-filter">
        <input type="radio" name="category" id="filter-all" checked>
        <label for="filter-all">All</label>

        <input type="radio" name="category" id="filter-headphones">
        <label for="filter-headphones">Headphones</label>

        <input type="radio" name="category" id="filter-watches">
        <label for="filter-watches">Watches</label>

        <input type="radio" name="category" id="filter-accessories">
        <label for="filter-accessories">Computer Accessories</label>

        <input type="radio" name="category" id="filter-power">
        <label for="filter-power">Power</label>
      </div>
```

css/products.css - Task 1 part (accent hue H = (61 x 7) mod 360 = 67)

```
/* Category Filter */
.category-filter {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin: 20px 0 25px;
}

.category-filter input {
  position: absolute;
  opacity: 0;
  width: 1px;
  height: 1px;
}

.category-filter label {
  display: inline-block;
  padding: 10px 18px;
  border: 1px solid var(--line);
  border-radius: 6px;
  cursor: pointer;
  background: var(--white);
  color: var(--ink);
}

/* Selected filter */
.category-filter input:checked + label {
  background: hsl(67, 70%, 45%);
  color: white;
  border-color: hsl(67, 70%, 45%);
}

/* Keyboard focus */
.category-filter input:focus-visible + label {
  outline: 2px solid hsl(67, 70%, 45%);
  outline-offset: 3px;
}

/* Hide products that do not match the selected category */
.products-section:has(#filter-headphones:checked)
.product-card:not(.headphones),
.products-section:has(#filter-watches:checked)
.product-card:not(.watches),
.products-section:has(#filter-accessories:checked)
.product-card:not(.accessories),
.products-section:has(#filter-power:checked)
.product-card:not(.power) {
  display: none !important;
}
```

Code - Task 2: Product page grid + sticky details box

product.html - new article structure (long description shortened here, full text in the file)

```
<article class="product-detail">

      <!-- Product Image -->
      <div class="product-image">
        <img src="./images/3.jpg" alt="Headphone Gen 2">
      </div>

      <!-- Product Details -->
      <div class="product-info">
        <h1>Headphone Gen 2</h1>

        <p class="detail-price">Rs. 3,499</p>

        <p class="product-summary">
          Experience crystal-clear sound with our latest headphones.
          Featuring active noise cancellation and a comfortable fit
          for extended listening sessions.
        </p>

        <ul class="spec-list">
          <li>Active noise cancellation</li>
          <li>40-hour battery life</li>
          <li>Quick charge: 10 minutes for 4 hours of playback</li>
          <li>Weight: 250 g</li>
        </ul>

        <div class="qty-row">
          <div class="qty-group">
            <label for="qty">Quantity</label>
            <input type="number" id="qty" name="qty"
                  min="1" max="10" value="1">
          </div>

          <button type="button" class="btn btn-cart">
            Add to Cart
          </button>
        </div>
      </div>

      <!-- Full-Width Description -->
      <section class="product-description">
        <h2>Product Description</h2>

        <p>
          The Headphone Gen 2 is designed for everyday listening,
          whether you are studying, working, travelling, or relaxing
          at home. Its comfortable ear cushions and adjustable
          headband help provide a secure fit during long sessions.
        </p>

        <p>
          Active noise cancellation helps reduce surrounding noise
          so you can concentrate on your music, lectures, podcasts,
          or online meetings. The built-in microphone also makes
          these headphones suitable for voice calls and virtual classes.
        </p>

        <!-- ... more description paragraphs and h3 sections (see product.html) ... -->
      </section>

    </article>
```

css/product-detail.css - grid areas (mobile first), then the BP rule with side-by-side areas and sticky

```
/* ===== Product Detail Layout ===== */

.product-detail {
  display: grid;
  grid-template-columns: 1fr;
  grid-template-areas:
    "image"
    "details"
    "description";
  gap: 2rem;
  align-items: start;
  margin: 2rem 0 4rem;
}

/* ===== Grid Areas ===== */

.product-image {
  grid-area: image;
}

.product-info {
  grid-area: details;
  min-width: 0;
}

.product-description {
  grid-area: description;
}

/* ===== Product Image ===== */

.product-image img {
  display: block;
  width: 100%;
  height: auto;
  border-radius: 0.75rem;
}

/* ===== Product Information ===== */

.product-info {
```

```
  padding: 1.5rem;
  border: 1px solid var(--line);
  border-radius: 0.75rem;
  background: var(--white);
}

.product-info h1 {
  margin-top: 0;
  margin-bottom: 1rem;
  font-size: 2rem;
}

.detail-price {
  margin: 0 0 1rem;
  font-size: 1.5rem;
  font-weight: 700;
  color: var(--accent-dark);
}

.product-summary {
  line-height: 1.7;
}

/* ===== Product Specifications ===== */

.spec-list {
  padding-left: 1.25rem;
  line-height: 1.9;
  margin: 1.5rem 0;
}

/* ===== Quantity and Cart ===== */

.qty-row {
  display: flex;
  flex-wrap: wrap;
  align-items: end;
  gap: 1rem;
  margin-top: 1.5rem;
}

.qty-group {
  display: flex;
  flex-direction: column;
  gap: 0.5rem;
}

.qty-group input {
  width: 90px;
  padding: 0.65rem;
  border: 1px solid var(--line);
  border-radius: 0.4rem;
  font: inherit;
}

.qty-row .btn {
  cursor: pointer;
}

/* ===== Full-Width Description ===== */

.product-description {
  line-height: 1.8;
}

.product-description h2 {
  margin-top: 0;
  margin-bottom: 1rem;
}

.product-description h3 {
  margin-top: 2rem;
  margin-bottom: 0.5rem;
}

.product-description p {
  margin-bottom: 1.25rem;
}

/* ===== Desktop Layout: BP = 661px ===== */

@media (min-width: 661px) {
  .product-detail {
    grid-template-columns: minmax(0, 1fr) minmax(0, 1fr);
    grid-template-areas:
      "image details"
      "description description";
    gap: 2rem;
  }

  .product-info {
    position: sticky;
    top: 91px; /* (N mod 40) + 70 = (61 mod 40) + 70 = 91px */
    align-self: start;
  }
}
```

Code - Task 3: JavaScript live search and price sort

index.html - search box, sort button, "No products found" message and script link

```
<!-- Search and Sort -->
      <div class="product-controls">

          <input type="text" id="search-input" placeholder="Search products...">
          <button type="button" id="sort-price">Sort by Price</button>

      </div>

      <!-- No Products Message -->
      <p id="no-products" class="no-products">
          No products found
      </p>

...

<script src="./script.js"></script>
```

script.js (new file)

```
// Get HTML elements
const searchInput = document.getElementById("search-input");
const sortButton = document.getElementById("sort-price");
const productGrid = document.querySelector(".product-grid");
const noProducts = document.getElementById("no-products");

// Get existing product cards
const products = document.querySelectorAll(".product-card");

// Premium price limit
const premiumLimit = 61 * 100;

// Sorting direction
let ascending = true;

// ================================
// 1. ADD PREMIUM LABELS
// ================================

products.forEach(function(product) {

    const priceText = product.querySelector(".product-price").textContent;

    const price = Number(priceText.replace(/[^0-9]/g, ""));

    if (price > premiumLimit) {

        const label = document.createElement("span");

        label.textContent = "Premium";
        label.className = "premium-label";

        product.querySelector(".card-details").appendChild(label);

    }

});

// ================================
// 2. SEARCH AND CATEGORY FILTER
// ================================

function filterProducts() {

    const searchText = searchInput.value.toLowerCase();

    let visibleCount = 0;

    products.forEach(function(product) {

        const name = product.querySelector(".product-name")
            .textContent.toLowerCase();

        const matchesSearch = name.includes(searchText);

        if (matchesSearch) {
            product.style.display = "";
            visibleCount++;
        } else {
            product.style.display = "none";
        }

    });

    // Check which products are actually visible after CSS filtering
    const actuallyVisible = Array.from(products).some(function(product) {
        return product.style.display !== "none" &&
               getComputedStyle(product).display !== "none";
    });

    if (!actuallyVisible) {
        productGrid.style.display = "none";
        noProducts.style.display = "block";
    } else {
        productGrid.style.display = "grid";
        noProducts.style.display = "none";
    }

}

// Search event
searchInput.addEventListener("input", filterProducts);

// Re-check the "No products found" message when the category changes
document.querySelectorAll('input[name="category"]').forEach(function(radio) {
    radio.addEventListener("change", filterProducts);
```

```
});

// ================================
// 3. SORT PRODUCTS BY PRICE
// ================================

sortButton.addEventListener("click", function() {

    // Convert existing product cards into an array
    const productArray = Array.from(products);

    // Sort according to price
    productArray.sort(function(a, b) {

        const priceA = Number(
            a.querySelector(".product-price").textContent.replace(/[^0-9]/g, "")
        );

        const priceB = Number(
            b.querySelector(".product-price").textContent.replace(/[^0-9]/g, "")
        );

        if (ascending) {

            return priceA - priceB;

        }

        else {

            return priceB - priceA;

        }

    });

    // Move existing cards using appendChild
    productArray.forEach(function(product) {

        productGrid.appendChild(product);

    });

    // Reverse sorting direction
    ascending = !ascending;

    // Apply search and category filters again
    filterProducts();

});

// Initial filtering
filterProducts();
```

css/products.css - Task 3 part

```
/* Search and Sort Controls */

.product-controls {
    display: flex;
    flex-wrap: wrap;
    gap: 12px;
    margin: 20px 0;
}

#search-input {
    flex: 1;
    min-width: 200px;
    padding: 12px;
    border: 1px solid var(--line);
    border-radius: 6px;
    font: inherit;
}

#sort-price {
    padding: 12px 18px;
    border: none;
    border-radius: 6px;
    background: var(--accent);
    color: white;
    cursor: pointer;
    font: inherit;
}

/* Premium Label */

.premium-label {
    display: inline-block;
    width: fit-content;
    margin-top: 10px;
    padding: 4px 10px;
    border-radius: 4px;
    background: hsl(67, 70%, 45%);
    color: white;
    font-size: 0.75rem;
    font-weight: 700;
}

/* No Products Message */

.no-products {
    display: none;
    text-align: center;
    padding: 30px;
    font-size: 1.2rem;
    color: var(--muted);
}
```

Extra: header fix for 375px (css/header.css - added at the end)

```
/* ===== Midterm: keep the header inside 375px screens ===== */
@media (max-width: 480px) {
  .site-nav ul {
    gap: 0.9rem;
  }

  .cart-link {
    padding: 0.4rem 0.7rem;
  }
}
```